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Integrating Multi-objective Optimization and Spatial Analysis for the Site Selection and Improvement of Levee Crossing Facilities

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Abstract

Levee systems protect riverfront areas from flooding but often create barriers that limit local mobility. In many communities, residents face restricted access to riverside parks and essential services due to a lack of crossing facilities. This study introduces an integrated framework that combines spatial analysis with multi-objective optimization to guide the siting and improvement of levee crossings. The framework evaluates five objectives: construction cost, sidewalk improvement cost, population coverage, spatial equity, and crash risk, using road networks, population data, and crash records. Two optimization methods are applied and compared: the Non-dominated Sorting Genetic Algorithm III (NSGA III) and Simulated Quantum Annealing (SQA). NSGA III tends to concentrate solutions within a specific region of the Pareto front, while SQA performs better in less-explored areas. SQA uses weight-driven scalarization to target specific planning priorities, while NSGA III supports broader exploration across objectives. Together, they form a complementary toolkit: NSGA III identifies promising trade-offs, and SQA refines solutions based on policy preferences. A simulation of 40 candidate sites reveals how cost, accessibility, equity, and safety interact, highlighting underserved and high-risk zones. This dual-method approach supports informed, equitable, and implementable infrastructure decisions in flood-prone urban environments.

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1. Introduction

Levee systems are engineered embankments designed to protect urban and rural areas from riverine and coastal flooding. By controlling water flow and preventing inundation, they safeguard infrastructure, property, and human lives. These structures are critical components of flood management strategies, providing resilience against extreme weather events and supporting the long-term sustainability of urban areas along rivers and coastlines (Smith et al., 2017). Beyond their protective function, levee systems interact with urban mobility networks, making facility siting and transportation planning essential for ensuring efficient access to public services. Many studies in this field prioritize accessibility metrics or shortest-path efficiency, focusing on minimizing travel distances and times to enhance overall transportation performance (Antonova et al., 2022; Glendinning et al., 2009; Starzyk et al., 2023). However, such approaches often fail to fully consider the multidimensional aspects of urban equity and safety, limiting their capacity to support inclusive and resilient urban mobility. In practice, connectivity between inner neighborhoods and riverside green spaces is frequently limited, and the uneven distribution of levee crossings results in accessibility gaps and inequitable use of public spaces (Papakonstantinou et al., 2019).

One potential solution is to employ an optimization framework that systematically evaluates candidate locations for levee crossings. Traditional optimization approaches rarely integrate geospatial data with comprehensive multi-objective formulations, which limits their ability to capture the complex interactions between urban infrastructure, population distribution, and mobility patterns (Rahman and Szabó, 2021). In the context of levee crossings, decision-making must account for multiple factors: construction feasibility, which ensures that the selected sites are technically and economically viable; service coverage, which measures how well the crossings connect neighborhoods to essential services and recreational areas; spatial equity, which addresses the fair distribution of accessibility across different communities; and traffic safety, which incorporates crash data to minimize risk for pedestrians and cyclists. Without such integrated tools, urban planners may inadvertently reinforce existing inequities, where certain communities remain underserved and vulnerable despite substantial infrastructure investment. By incorporating multiple objectives and spatial analysis into the decision-making process, an optimization framework can provide a more balanced and data-driven approach to improving connectivity, accessibility, and safety in flood-prone urban areas.

Evolutionary algorithms have been widely applied to facility siting, environmental management, and resource allocation. Among these, the Non-dominated Sorting Genetic Algorithm II (NSGA-II) (Deb et al., 2002) and its extension NSGA-III (Deb & Jain, 2014) are particularly effective for multi-objective optimization problems. These algorithms generate a set of compromise solutions that balance competing objectives, rather than producing a single optimal solution, which allows planners to evaluate trade-offs between factors such as cost, accessibility, equity, and safety. NSGA-III extends the capabilities of NSGA-II to handle problems with more than three objectives, employing advanced non-dominated ranking and diversity-preserving mechanisms to maintain a well-distributed set of solutions across the objective space. By using these evolutionary algorithms, decision-makers can systematically explore multiple scenarios and identify solutions that achieve the best possible balance among competing priorities in complex urban infrastructure planning.

In addition to evolutionary algorithms, quantum-inspired optimization methods have emerged to tackle large-scale combinatorial problems that are difficult to solve using traditional techniques. Simulated Quantum Annealing (SQA), formulated within a Quadratic Unconstrained Binary Optimization (QUBO) framework, explores complex energy landscapes by mimicking quantum tunnelling effects, which allows it to escape local optima and search more efficiently for global solutions. While originally developed for quantum hardware, SQA can also be implemented on classical and hybrid computing systems, making it a practical tool for high-dimensional, discrete decision problems. In the context of levee crossing planning, SQA has the potential to efficiently evaluate large numbers of candidate locations while simultaneously considering multiple objectives such as cost, coverage, equity, and safety, complementing traditional evolutionary approaches and offering scalable solutions for complex urban infrastructure challenges.

Therefore, this study proposes an integrated framework that combines spatial analysis with multi-objective optimization for the site selection of levee crossing facilities. The framework utilizes geospatial data, including road

networks, population distributions, and crash datasets. Two optimization strategies are employed: NSGA-III, representing classical evolutionary algorithms, and SQA, executed on a specialized hardware platform provided by Compal Electronics. Both methods are applied to improve the siting of levee crossings, and their results are compared to demonstrate how advanced optimization techniques can be tailored to complex geospatial problems. The framework addresses multiple objectives, including minimizing construction and sidewalk improvement costs, maximizing population coverage, enhancing spatial equity, and reducing crash risk. It is applied to Sanchong and Luzhou districts in New Taipei City, Taiwan, which are densely populated areas adjacent to the levee and illustrate the challenges of limited connectivity.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on levee facilities, transport equity, and multi-objective optimization. Section 3 presents the study area and data sources then followed by the methodological framework and optimization models. Section 4 reports the results and discusses their implications. Section 5 concludes with key findings, policy relevance, and directions for future research.

2. Literature Review

2.1. Flood-Control Infrastructure and Urban Mobility

Flood-control infrastructure is widely recognized as a critical component of urban resilience, designed to protect settlements, infrastructure, and communities from the increasing threats of sea-level rise and riverine flooding. Levees, in particular, are engineered to regulate water flow and prevent inundations, thereby mitigating potential damage to property and human life. While their primary function is hydrological, a growing body of research highlights that these structures also have significant impacts on urban transport networks. By acting as physical barriers along riverfronts, levees can hinder movement, limit connectivity, and reduce accessibility to public services and recreational spaces.

Several studies have investigated how flood-control infrastructure affects both protection and mobility. Papakonstantinou et al. (2019), for example, analyzed the San Francisco Bay Area using a combination of hydrodynamic flood modeling and traffic assignment. They demonstrated that prioritizing shoreline protection without considering mobility impacts can generate additional congestion and disrupt traffic flow. Their results also suggest that providing full flood protection is not always the most cost-effective approach, highlighting the need to balance flood mitigation with efficient transportation. Similarly, Mitropoulos et al. (2025) examined accessibility under various flood scenarios and showed that even moderate flooding can substantially restrict access to essential facilities, including hospitals and schools. These studies collectively demonstrate that flood events can increase vulnerability for populations that rely on safe and reliable pedestrian and vehicular access.

Together, these studies emphasize that flood-control structures should be recognized not only as protective barriers but also as key factors shaping spatial accessibility and urban mobility. They underscore the importance of integrated planning approaches that consider flood protection, infrastructure siting, and equitable access simultaneously. In this context, understanding the interactions between levees, transportation networks, and population distribution is essential for developing strategies that strengthen urban resilience while maintaining accessibility for all residents.

2.2. Facility location and multi-objective optimization

Infrastructure siting problems are inherently complex and multi-dimensional, requiring careful consideration of multiple factors such as cost, coverage, and resilience. Early models primarily focused on minimizing travel distances or construction costs, often relying on simple mathematical formulations that did not capture the interactions between population distribution, accessibility, and infrastructure constraints such as existing road networks or land use patterns. Advances in evolutionary algorithms have enabled more sophisticated approaches that can simultaneously address multiple variables and competing objectives. For example, Deb et al. (2002) introduced the NSGA-II evolutionary algorithm, which generates a set of Pareto-optimal solutions, providing decision-makers with a spectrum of alternatives that balance competing goals rather than a single “best” solution. This flexibility allows planners to evaluate trade-offs, such as choosing between lower costs or higher service coverage. Building on this, Deb and Jain

(2014) developed NSGA-III to handle many-objective problems by incorporating reference-point guided Pareto exploration. Unlike NSGA-II, which is best suited for two to three objectives, NSGA-III can efficiently manage scenarios with five or more objectives, enhancing solution diversity in high-dimensional decision spaces. Despite their widespread application, these algorithms have largely remained within classical evolutionary frameworks, and few studies have applied them to levee-related infrastructure planning, where flood protection, mobility, equity, and safety must be considered together.

Recent applications demonstrate the adaptability and potential of these multi-objective optimization approaches across diverse domains. Zhang et al. (2025), for example, applied U-NSGA-III to design urban air mobility corridors in Chengdu, simultaneously considering demand, safety, and cost. By generating multiple Pareto-optimal solutions, the study enabled planners to evaluate trade-offs between operational efficiency and safety risk under uncertain demand conditions, helping to identify corridor designs that balance service coverage with acceptable risk levels. However, the methods employed in this context are not directly transferable to pedestrian-scale infrastructure siting, which requires fine-grained spatial analysis that accounts for local connectivity, walking distances, street-level accessibility, and the distribution of vulnerable users.

Similarly, Maliki et al. (2025) focused on mobile facility placement under environmental and disruption risks using a multi-period optimization framework that balances resilience and cost efficiency. While effective for dynamic, large-scale problems, this framework did not explicitly incorporate equity or user safety considerations, which are essential when planning facilities that serve pedestrians or other vulnerable populations. In the healthcare sector, Emergency Medical Services (EMS) facility planning has also employed multi-objective optimization, integrating factors such as travel time, demographic vulnerability, and traffic risk (Strauss et al., 2024). These applications demonstrate the value of optimization in balancing multiple objectives, but they are primarily tailored to emergency response scenarios and are not designed to address persistent physical barriers, such as levees, that limit daily pedestrian and cyclist accessibility.

In parallel, quantum-inspired optimization methods have emerged as a promising frontier for large-scale combinatorial siting problems. SQA explores complex solution landscapes by mimicking the behavior of quantum particles as they tunnel through energy barriers. This imitation allows the algorithm to escape local optima, which are suboptimal solutions that traditional methods may get stuck in, and more efficiently search for global optimal solutions in high-dimensional and discrete decision spaces. This capability is particularly valuable for high-dimensional and discrete decision problems, where conventional evolutionary algorithms may struggle to explore the full solution space efficiently. Although SQA applications in transportation and urban infrastructure are still limited, combining this approach with GIS-based datasets provides new opportunities for addressing complex planning problems. In the context of levee crossings, SQA offers the potential to evaluate numerous candidate sites while simultaneously optimizing multiple objectives, including cost, coverage, equity, and safety, thereby complementing classical evolutionary methods and providing scalable solutions for urban infrastructure planning.

2.3. Transport equity and safety

Equity has emerged as a central principle in transport planning, emphasizing not only efficiency but also fairness in access across different social groups. Litman (2020) distinguishes between horizontal equity, which ensures equal treatment of users with similar needs, and vertical equity, which prioritizes the needs of disadvantaged populations. These concepts have been operationalized in studies assessing accessibility, service coverage, and affordability. For instance, Böcker et al. (2019) compared daily mobility patterns across Dutch, Norwegian, and Swedish cities, highlighting how environmental factors, such as weather, disproportionately affect different population groups. Safety is another critical dimension of transport equity. Sabi Boun et al. (2024) reviewed pedestrian safety interventions in low- and middle-income countries and concluded that measures such as improved crossings and enhanced visibility are among the most effective strategies to reduce accident risks. Similarly, research on urban roadway design has shown that infrastructure supporting safe walking and cycling improves accessibility for vulnerable populations, including children and the elderly. The EMS optimization study further demonstrated that integrating demographic

vulnerability with traffic risk data enhances equity in emergency response outcomes (Strauss et al., 2024). Collectively, these findings underscore that transport equity is not merely a theoretical concept but a practical concern that directly shapes everyday accessibility and safety.

3. Study Data and Methodology

3.1. Study area & Data

The study area in Fig. 1 includes Sanchong and Luzhou Districts in New Taipei City, Taiwan, located along the western bank of the Tamsui River. Together, these districts are among the most densely populated in the region, with a combined population exceeding 800,000. The urban environment is characterized by compact residential neighborhoods, mixed-use developments, and a well-connected street network. Although riverside parks and recreational spaces are available, access is limited by levee structures, with crossing facilities that are unevenly distributed. This combination of high population density, constrained mobility, and barriers to riverside access makes Sanchong and Luzhou a representative case for examining levee crossing planning, particularly regarding resilience, equity, and pedestrian safety.

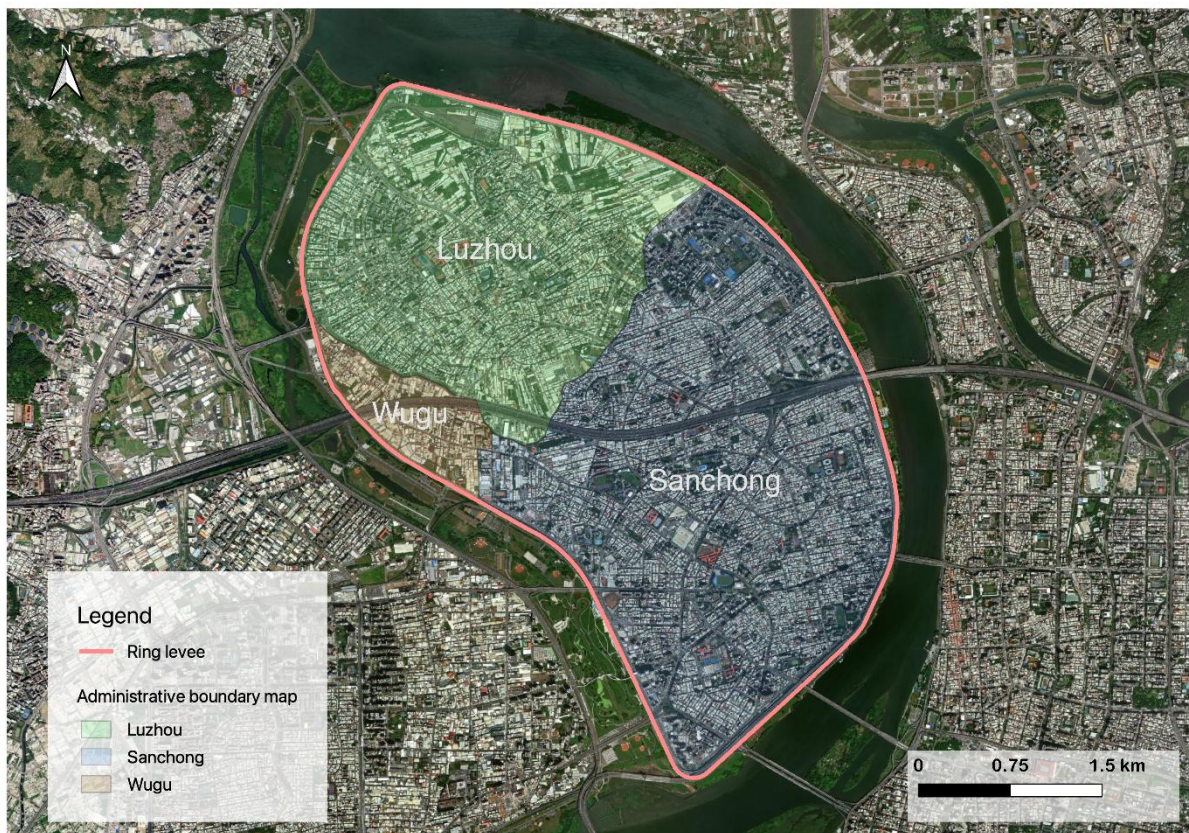


Fig. 1. The study area of Luzhou and Sanchong District in New Taipei City, Taiwan

For this study, seven datasets were compiled for analysis: (i) levee geometry, (ii) population statistics, (iii) village boundary polygons, (iv) road network data, (v) sidewalk inventories, (vi) existing levee crossing facilities, and (vii) crash records. These datasets were obtained from the New Taipei City Government, the National Land Surveying and Mapping Center, the National Police Agency, and OpenStreetMap. The data were integrated into a pedestrian demand network, which served as the basis for spatial analyses. This process enabled the evaluation of shortest walking paths

as well as the identification of pedestrian demand intensity and pressure points along the levee. The overall process of this study is summarized in Fig. 2, while the detailed procedures are described in the following sections.

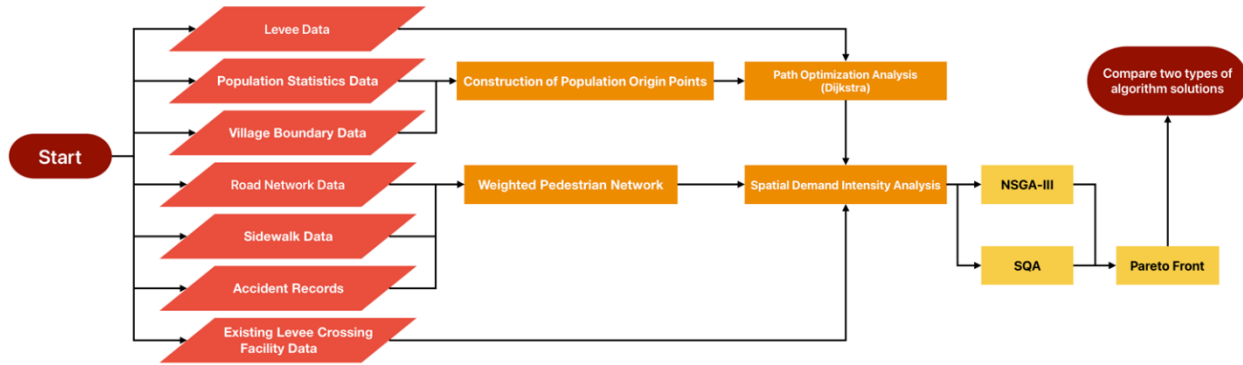


Fig. 2. Flow chart.

3.2. Data Processing

In terms of data and spatial object construction, this study first converted village-level population statistics into demand source data. Specifically, population counts for each village were transformed into randomly distributed points within administrative boundaries, which served as residential origins. To reduce computational complexity, a 10% sample of these points was selected for analysis. Each point was then aligned with the nearest road network, node or edge, ensuring that pedestrian demand was accurately linked to the actual walking infrastructure rather than existing as abstract locations in space.

The ring levee was modeled as a closed boundary, functioning as the outer edge of the pedestrian network and representing the point at which all walking routes terminate. In other words, the levee acts as a barrier: pedestrians can approach it but cannot cross freely except at designated crossing facilities. Conceptually, this makes the levee the “end point” of accessibility for residents unless a crossing exists. While, in theory, each origin could connect to any location along the levee, Dijkstra’s shortest-path algorithm was applied to calculate walking routes in practice. The algorithm identifies the nearest projected location on the levee from each origin, which is then defined as the effective destination. This setup allows the model to simulate realistic walking demands toward the levee boundary while also highlighting how accessibility depends on the presence or absence of crossing facilities.

During route computation, in addition to measuring short distance, the model also calculated the total length of road segments lacking sidewalks. When multiple routes passed through the same “missing sidewalk” segment, the segment was counted only once, since sidewalk improvements are one-time infrastructure costs that do not scale with the number of users. Furthermore, accident risk values were ultimately embedded into the shortest-path layer to represent pedestrian exposure along walking routes, while the specific methods for calculating sidewalk improvement costs and accident risk are presented in detail in Section 3.3.

After completing the route computation, the study further developed the candidate facility layer, consisting of 103 predefined potential crossing points along the levee line as shown in Fig. 3. In addition, for subsequent analysis, a 103×103 distance coefficient matrix was generated to capture pairwise distances between all candidate sites. It is important to emphasize that this matrix is not a GIS layer but serves as an essential input for optimization, with its detailed application discussed later in Section 3.3.3.

At this stage, three GIS-based layers were established: (1) the candidate facility layer, containing all predefined crossing sites with their construction costs; (2) the accident layer, produced by smoothing and processing the distribution of traffic accidents to represent risk exposure; and (3) the population heatmap layer, generated by integrating residential origins with the pedestrian network to capture demand distribution around the levee. Together, these layers provided the foundation for subsequent multi-objective optimization, as illustrated in Fig. 4.

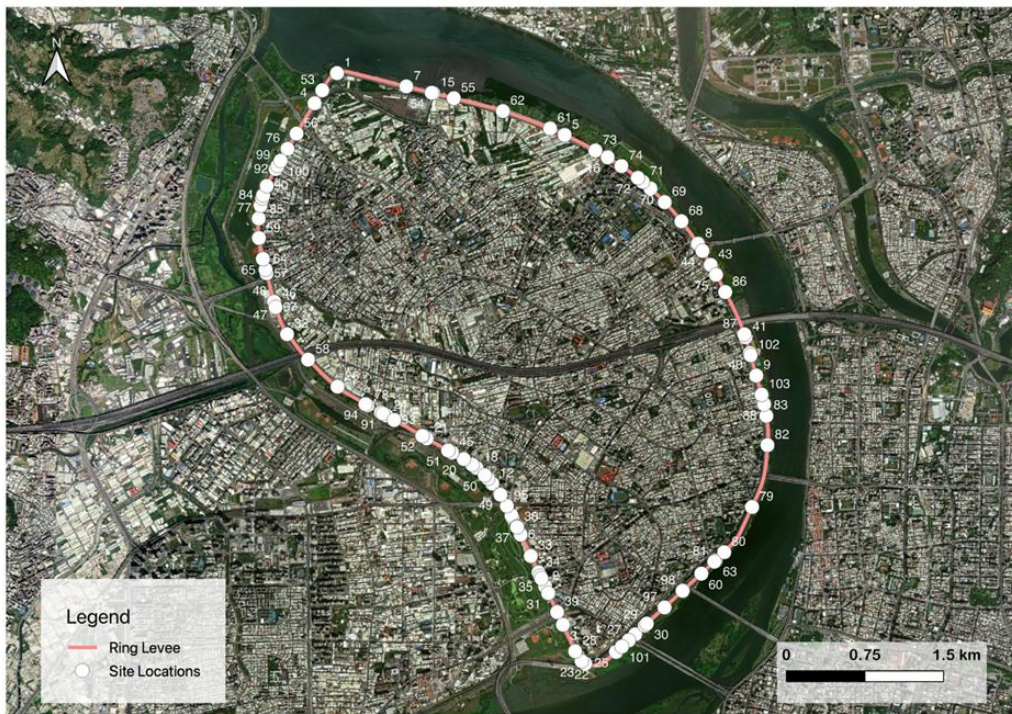


Fig. 3. Predefined potential crossing points along the levee line GIS Data Processing Map

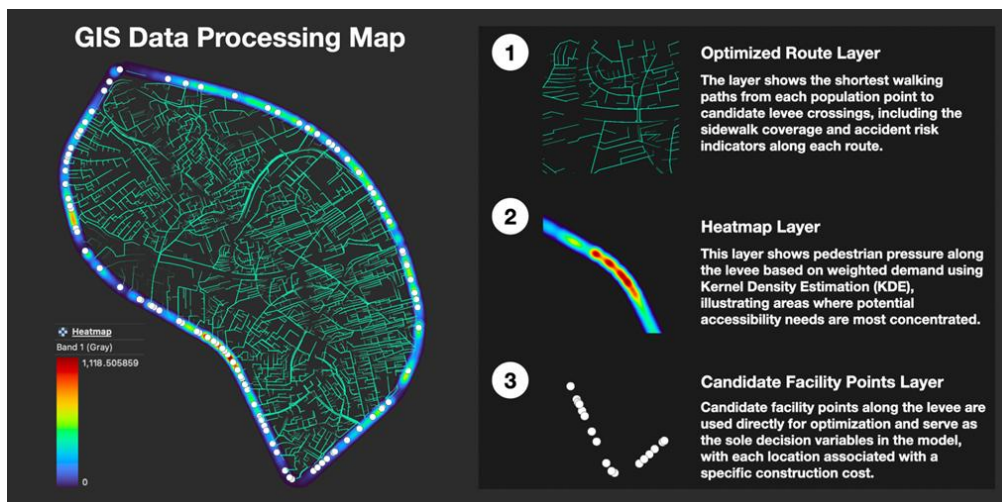


Fig.4. GIS Data Processing Map

Finally, the shortest-path destinations were systematically embedded into the candidate facility dataset. Concretely, each shortest path terminated at a projected point on the levee, and this destination was then snapped to the nearest candidate facility among the 103 predefined sites. Through this matching procedure, the cumulative attributes associated with each path, such as population counts, sidewalk improvement costs, and accident exposures, were aggregated and assigned to the corresponding candidate site. As a result, every candidate location was enriched with four categories of information: (1) population demand, represented by the number of origins whose shortest paths terminated at that site; (2) sidewalk improvement cost, counted as a fixed one-time expense derived from the

aggregated missing-sidewalk segments along those paths; (3) accident risk exposure, expressed as the cumulative crash counts from paths leading to the site; and (4) construction cost, initialized to a baseline value of 1. Together with the inter-candidate distance matrix, this embedding process completed the dataset preparation for subsequent multi-objective optimization.

3.3. Optimization Model

The proposed optimization model incorporates five key objectives: minimizing construction cost, minimizing sidewalk improvement cost, maximizing population coverage, enhancing spatial equity, and reducing accident risk. To identify and evaluate Pareto-optimal solutions, the model employs two complementary solution techniques: Non-dominated Sorting Genetic Algorithm III (NSGA-III) and Simulated Quantum Annealing (SQA).

3.3.1. Demand Estimation

Population demand is estimated using a continuous density surface generated through KDE, which helps identify areas with high potential need for levee crossings. Residential origin points, weighted by population size, are mapped onto the pedestrian road network to facilitate accurate calculation of the shortest path distances to potential levee crossing sites. A Gaussian kernel function is then applied to these points to create a smoothed “population heat map” that reflects the intensity of demand across space. The demand score p_i for each candidate crossing site is calculated based on the KDE intensity at its location, using the formula (Silverman, 1986) as follows.

$$p_i = \sum_{j=1}^M W_j \cdot \exp\left(-\frac{d_{ij}^2}{2h^2}\right) \quad (1)$$

Where d_{ij} is the network-based distance between origin j and candidate site i , h is the bandwidth parameter controlling the spread of demand influence, W_j is the population weight at origin j , and M is the total number of residential origin points included in the analysis. Equation (1) captures not only the presence of nearby residents but also the gradual decay of service potential with increasing distance, providing a continuous measure of accessibility demand across the levee.

3.3.2. Accident Risk Assessment

Accident risk is assessed by combining network-based kernel density estimation (NKDE) with shortest path analysis to capture both the spatial distribution of past accidents and the actual walking routes pedestrians would take. First, historical accident locations are smoothed along the pedestrian network using NKDE, producing a continuous risk intensity value $H(e)$ for each network edge e . This reflects how accident-prone each segment of the network is. Then, for every pair of residential origin j and candidate crossing site i , the total accident exposure along the shortest walking path $SP(j, i)$ is calculated by summing the risk values of all edges along the path, weighted by their lengths using the followed formula (Dijkstra (1959); Xie & Yan (2008)):

$$PR_{ji} = \sum_{e \in SP(j,i)} H(e)L_e \quad (2)$$

Here, L_e is the length of edge e . To evaluate the overall risk associated with a candidate site i , the model aggregates the exposure values from all origins, weighting each by its population size W_j , and then applies normalization to ensure comparability across sites:

$$r_i = \text{norm}(\sum_{j \in J} W_j P R_{ji}) \quad (3)$$

Equation (3) modified from Xie & Yan (2008). The original paper defines accident risk exposure through NKDE, but Equation (3) introduces a normalization step to rescale the weighted risk values into $[0,1]$ for comparability across objectives. This approach captures the relative safety of walking routes to each candidate crossing. Sites that require pedestrians to traverse more dangerous paths, those with higher accident densities, receive higher risk scores, which are penalized in the optimization process to promote safer access.

3.3.3 Spatial Equity (Dispersion Penalty)

To encourage a fair and balanced distribution of levee crossings, the model incorporates a dispersion penalty that discourages clustering of selected sites. This is achieved by penalizing candidate locations that are placed too close to one another along the levee. The penalty function is defined as:

$$E(x) = \sum_{i \neq j} \exp(-\lambda d_{ij}^{\text{levee}}) x_i x_j \quad (4)$$

In Equation (4), d_{ij}^{levee} represents the distance between candidate sites i and j along the levee, $x_i x_j$ are binary decision variables indicating whether each site is selected, and λ is a decay parameter controlling the strength of the penalty. Equation (4) adopted from the p-dispersion problem. The original works (Kuby 1987; Erkut 1990) define dispersion in terms of maximizing minimum distances or maximizing summed distances between selected facilities. Equation (4) adapts this concept by applying an exponential distance-decay function, penalizing facility pairs that are too close along the levee. By assigning higher penalty values to closely spaced sites, the model promotes a more dispersed and equitable spatial layout of crossings, ultimately improving accessibility for residents across the entire study area.

3.3.4 Objectives

The optimization model balances multiple planning goals by integrating five distinct objectives related to benefit, cost, equity, and safety. These objectives guide the selection of levee crossing sites to ensure that the final solution is both effective and equitable.

- Objective 1: Maximize population coverage

$$\min F_1(x) = - \sum_{i \in I} p_i x_i \quad (5)$$

This objective seeks to maximize accessibility for nearby residents. Adopted from Church & Reville (1974), p_i represents the demand score at candidate site i , derived from KDE, and x_i is a binary variable indicating whether site i is selected (1) or not (0). The negative sign ensures that higher coverage leads to a lower objective value, aligning with the minimization framework.

- Objective 2: Minimize sidewalk improvement cost

$$\min F_2(x) = \sum_{i \in I} s_i x_i \quad (6)$$

This objective reduces the cost of upgrading pedestrian infrastructure. Equation (6) is modified from Daskin (2013). s_i denotes the estimated sidewalk improvement cost for candidate site i , and x_i again indicates site selection.

- Objective 3: Minimize construction cost

$$\min F_3(x) = \sum_{i \in I} c_i x_i \quad (7)$$

This objective controls the financial burden of building new crossings. c_i is the construction cost for site i , which is treated as a unit constant (i.e., $c_i = 1$) for all sites. An identical baseline value of 1 was assigned to all candidate facilities to maintain model consistency. As a result, this objective effectively minimizes the total number of selected sites. Equation (7) is also modified from Daskin (2013).

- Objective 4: Enhance spatial equity

$$\min F_4(x) = E(x) \quad (8)$$

This objective discourages clustering of facilities by applying a dispersion penalty. The function $E(x)$ penalizes pairs of selected sites that are located close to each other along the levee, promoting a more balanced spatial distribution. Equation (7) is also modified from Kuby (1987) and Erkut (1990) related the p-dispersion problem but extended by introducing an exponential decay to ensure facility separation is continuously penalized rather than discretely maximized.

- Objective 5: Minimize accident risk

$$\min F_5(x) = \sum_{i \in I} r_i x_i \quad (9)$$

This objective prioritizes pedestrian safety by minimizing exposure to hazardous walking routes. r_i is the normalized accident risk score for site i , calculated based on cumulative crash exposure along the shortest paths from residential origins. Equation (9) is modified from Xie & Yan (2008). While NKDE defines accident intensity along network edges, Equation (9) aggregates and normalizes exposure values into a site-level risk index, which is then minimized as part of the optimization.

To ensure comparability across different scales, all objectives except construction cost are normalized to the range [0,1]. This allows the optimization algorithms to evaluate trade-offs fairly across diverse criteria.

3.4. Optimization Algorithm

To solve the multi-objective facility location problem, two complementary algorithmic approaches are employed. The first is the Non-dominated Sorting Genetic Algorithm III (NSGA-III), a widely used evolutionary algorithm known for its effectiveness in handling many-objective optimization problems. NSGA-III uses a combination of reference point guidance and non-dominated sorting to generate a diverse set of solutions that approximate the Pareto front, representing trade-offs among competing objectives such as cost, coverage, equity, and safety.

The algorithm operates over a series of generations, where each generation represents one iteration of solution evaluation, selection, and reproduction. In this study, NSGA-III was executed for 2,000 generations, allowing the algorithm to transition from broad exploration of the solution space in early stages to refined convergence in later stages. This iterative process ensures that the final population includes well-distributed, high-quality solutions that reflect balanced compromises across all objectives.

The second approach reformulates the problem as a Quadratic Unconstrained Binary Optimization (QUBO) model and solves it using Simulated Quantum Annealing (SQA) on specialized quantum-inspired hardware. In this

framework, QUBO serves as the mathematical foundation for encoding the optimization problem into a binary format, where the objective function is expressed as a quadratic polynomial over binary variables. SQA then explores the solution space by simulating quantum tunneling dynamics, efficiently searching for low-energy configurations that correspond to optimal or near-optimal solutions. Unlike NSGA-III, which can directly optimize multiple objectives simultaneously, SQA requires scalarization: each objective is first normalized and then combined into a single weighted objective function. The scalarized function is defined as:

$$\min H = w_1 F_1(x) + w_2 F_2(x) + w_3 F_3(x) + w_4 F_4(x) + w_5 F_5(x) \quad (10)$$

In Equation (10): $F_1(x)$ through $F_5(x)$ represent the five individual objectives: population coverage, sidewalk improvement cost, construction cost, spatial equity, and accident risk. w_1 through w_5 are the corresponding weights assigned to each objective, reflecting their relative importance in a given scenario. x is the binary decision vector indicating which candidate sites are selected. Equation (10) adopted from the weighted-sum method in multi-objective optimization. Classical references define the weighted aggregation of objectives (Marler & Arora, 2004), but equation (10) adapts it specifically for SQA by normalizing each objective and systematically varying weights (w_1, w_2, w_3, w_4, w_5) to approximate the Pareto front.

This dual-method framework enables a robust comparison between a state-of-the-art evolutionary algorithm and a quantum-inspired metaheuristic. The analysis not only evaluates the relative performance of both solvers but also demonstrates the flexibility and resilience of the proposed modelling framework across different optimization paradigms.

4. Results and Discussion

4.1. NSGA-III search dynamics & convergence

Understanding the search dynamics of multi-objective optimization algorithms is essential for interpreting solution quality and trade-off behavior. The NSGA-III is particularly well-suited for many-objective problems due to its reference-point-based selection mechanism, which promotes both convergence toward the Pareto front and diversity across solutions. To visualize the algorithm's progression, Fig.5 presents a parallel-coordinates plot of NSGA-III's behavior over 2,000 generations, capturing how solutions evolve across five objectives: population coverage (F_1), sidewalk improvement cost (F_2), construction cost (F_3), spatial equity (F_4), and accident risk (F_5).

Each line in the plot represents an individual solution spanning the five objectives, while the color gradient from dark purple to bright yellow indicates the generation in which the solution was produced. Early generations (dark lines) exhibit a widespread across all objective axes, reflecting the algorithm's initial global exploration phase. This broad search is intentional, allowing NSGA-III to sample diverse regions of the solution space and avoid premature convergence. As the algorithm progresses, the population begins to converge toward more refined trade-offs. In later generations (bright yellow lines), solutions cluster within narrower ranges, indicating a shift toward local exploitation guided by reference points.

By the final stages of the search, NSGA-III converges toward a set of compromise solutions that reflect balanced trade-offs across all five objectives. Specifically, the selected solutions fall within the following ranges: population coverage F_1 is maximized, with values approximately between -26 and -20 ; sidewalk improvement cost F_2 ranges from 14 to 19 , indicating moderate infrastructure investment; construction cost F_3 , represented by the number of selected sites, spans from 55 to 74 ; spatial dispersion penalty F_4 falls between 200 and 300 , suggesting a reasonably balanced distribution of facilities along the levee; and accident risk exposure F_5 ranges from 11 to 15 , reflecting efforts to minimize pedestrian vulnerability. These ranges demonstrate the algorithm's capacity to balance competing objectives rather than minimizing each in isolation. For example, relatively higher values in F_2 through F_5 may still be considered optimal when evaluated in conjunction with strong performance in population coverage and construction cost, underscoring the multi-objective nature of the problem and the importance of compromise solutions.

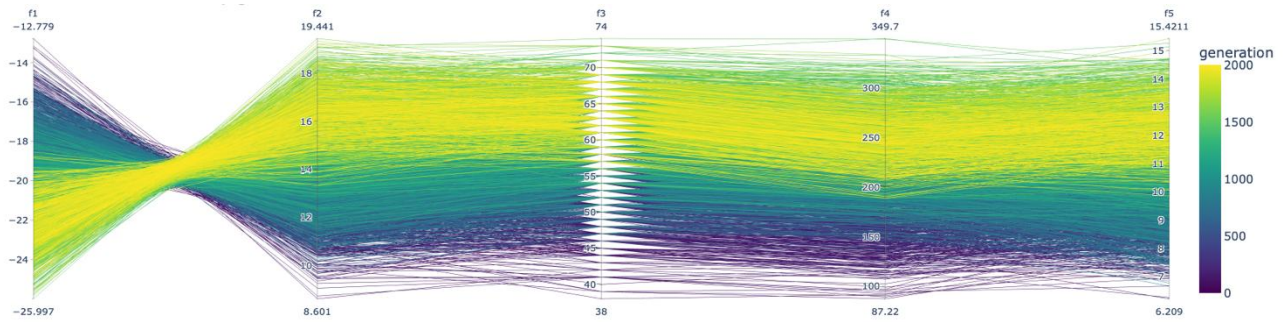


Fig. 5. NSGA-III Solutions (colored by generation)

Previous studies have further explored and enhanced NSGA-III's performance, particularly in maintaining a balance between convergence toward the Pareto front and preserving diversity among solutions. For instance, Gu et al. (2022) proposed an improved NSGA-III algorithm based on distance-based dominance to address the challenge of diversity maintenance in high-dimensional objective spaces. In standard NSGA-III, the reference-point mechanism guides the population toward evenly distributed solutions, but in very high-dimensional problems, solutions can still cluster unevenly, leaving regions of the objective space underrepresented. By incorporating distance-based dominance, the improved algorithm explicitly considers the spacing of solutions relative to one another, which encourages a more uniform spread across the Pareto front. This is particularly relevant to the early generations in Fig. 5, where NSGA-III is exploring broadly across all objectives.

In addition, Geng et al. (2023) introduced a dual-population-based NSGA-III (DP-NSGA-III), in which two populations evolve simultaneously and exchange offspring to enhance convergence and diversity. One population focuses on exploration while the other emphasizes exploitation, creating a synergistic effect that improves the algorithm's ability to discover high-quality solutions. This approach aligns with the later generations observed in Fig. 5, where bright-colored lines indicate convergence to specific regions of the objective space while still maintaining diversity across trade-offs.

Furthermore, Xu et al. (2024) developed a comprehensive adaptive penalty scheme applied to NSGA-III (caps-NSGA-III) to achieve a more balanced performance between convergence and diversity. Traditional penalty mechanisms in constrained optimization can overly favor feasibility, sometimes at the cost of exploring diverse regions. The caps-NSGA-III method dynamically adjusts penalties based on the population's progress, allowing the algorithm to maintain diversity while still converging effectively toward optimal trade-offs. This refinement explains how NSGA-III can navigate complex, constrain many-objective problems and achieve the compromise solutions in the ranges observed in Fig. 5 (e.g., $F_1 \approx -26$ to -20 , $F_4 \approx 200$ to 300).

Collectively, these studies highlight ongoing efforts to refine NSGA-III for complex many-objective problems. They demonstrate that enhancements to diversity preservation, population management, and adaptive constraint handling can significantly improve both the convergence and distribution of solutions. Fig. 5 exemplifies these principles, showing the transition from a broad exploratory search in early generations to a well-converged, balanced set of compromise solutions, reflecting the algorithm's capability to handle trade-offs among multiple conflicting objectives.

4.2. Comparative Search Behaviors of NSGA-III and SQA

In multi-objective optimization, the effectiveness of a solution method depends not only on its final output but also on how it navigates the solution space. Different algorithms employ distinct search strategies, which influence their ability to balance convergence, diversity, and computational efficiency. This section compares the search behaviours of two complementary approaches: NSGA-III and SQA. While NSGA-III is a population-based evolutionary algorithm that emphasizes diversity through reference-point guidance, SQA is a probabilistic metaheuristic inspired

by quantum annealing principles, favoring deep exploitation of promising regions.

Understanding how these algorithms explore, refine, and converge toward Pareto-optimal solutions is critical for interpreting their performance across competing objectives such as coverage, cost, equity, and safety. Through visualizations including parallel coordinates plots and pair grid overlay, this section examines the dynamic search trajectories of NSGA-III and SQA, highlighting their respective strengths and limitations in handling the complexity of the levee crossing facility location problem.

Fig. 6 through Fig. 8 illustrate the contrasting search dynamics of NSGA-III and SQA in generating Pareto-optimal solutions for the multi-objective facility location problem. NSGA-III exhibits a clear convergence pattern over time, particularly evident in objective F_3 , where the majority of solutions concentrate within the range of 45 to 52 selected sites. This convergence occurs as earlier, lower-quality solutions are progressively replaced by newer, more refined ones. The algorithm's reference-point mechanism intensifies exploration in promising regions, resulting in denser solution coverage where trade-offs are more favorable, while sparsely explored areas yield fewer solutions. Fig. 6 captures this progression, showing NSGA-III's transition from broad, global exploration in early generations to a focused search within a narrower band of high-quality compromises.

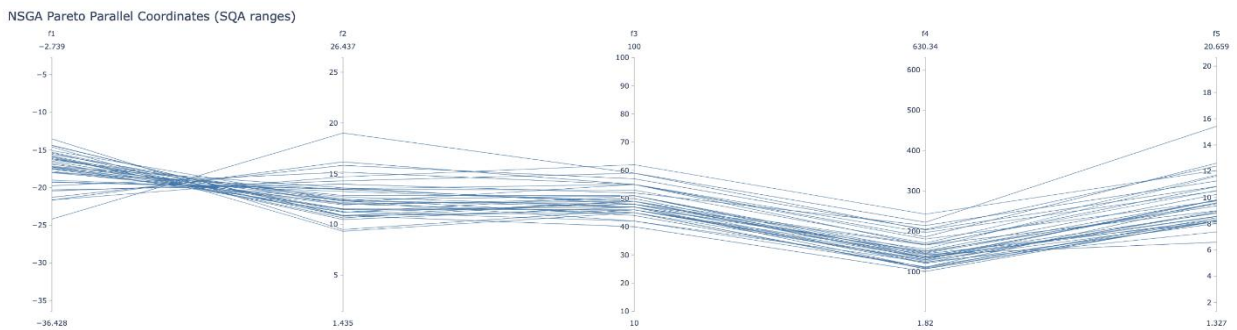


Fig. 6. NSGA-III Pareto Front Parallel Coordinates

In contrast to NSGA-III's convergence-driven search behavior, SQA produces a more evenly distributed set of solutions across the objective space. This pattern arises from SQA's design, which evaluates every 10 candidate sites as discrete breakpoints, enabling systematic sampling across a wide range of configurations. As a result, SQA achieves broader coverage of the solution space, with solutions spanning nearly all intervals of the defined objective ranges. Fig. 7 illustrates this weight-driven approach, where solutions are generated under specific weighting and cardinality constraints. Unlike population-based algorithms, SQA does not rely on evolutionary operators but instead uses probabilistic tunneling to escape local optima and explore alternative configurations. This mechanism contributes to its ability to maintain diversity and avoid premature convergence, making it particularly effective for identifying underrepresented trade-offs in high-dimensional optimization problems. This aligns with previous findings that highlight the value of weight-driven or breakpoint-based strategies in ensuring diversity and interpretability across the solution space (Kumar et al., 2025).

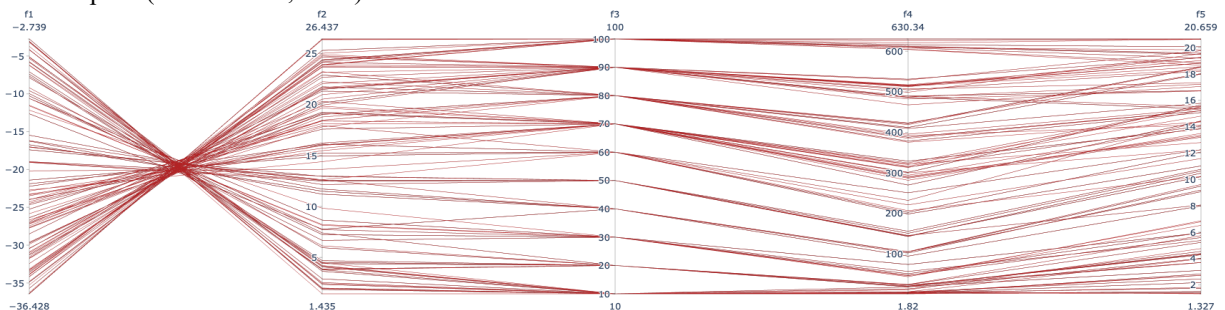


Fig. 7. SQA Solutions Parallel Coordinates

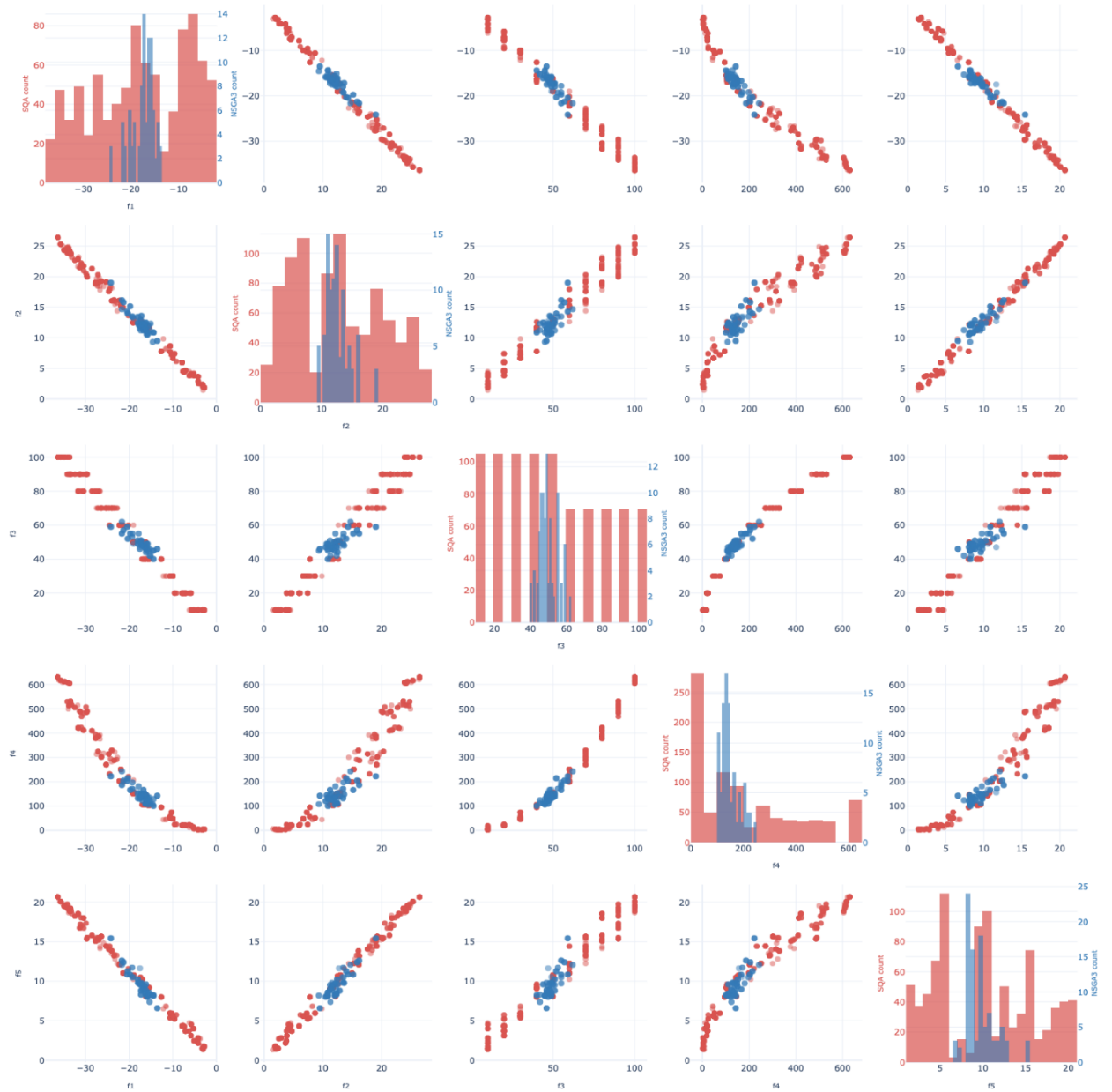


Fig. 8. Pair grid overlay: SQA (red), NSGA (blue)

Fig. 8 presents a combined visualization of solution distributions generated by NSGA-III and SQA, highlighting their distinct search behaviors within the multi-objective solution space. The blue points, representing NSGA-III outputs, are tightly clustered within a focused region of the objective space, reflecting the algorithm's convergence toward a narrow band of high-quality trade-offs. This clustering is a result of NSGA-III's reference-point-guided selection mechanism, which intensifies search efforts around promising regions as generations progress. In contrast, the red points, corresponding to SQA-generated solutions, exhibit a broader and more dispersed distribution. SQA not only covers the region explored by NSGA-III but also extends into more extreme or diverse trade-off zones, capturing

configurations that may be underrepresented in evolutionary search. This contrast underscores the complementary nature of the two algorithms: NSGA-III excels in refining and converging toward balanced solutions, while SQA contributes exploratory depth and diversity, offering planners a wider spectrum of alternatives for decision-making under competing objectives.

4.3. Pairwise Dominance Analysis

NSGA-III and SQA demonstrate bidirectional dominance, meaning that neither algorithm is consistently better than the other across all objectives; instead, each has areas of strength depending on the solution characteristics. To quantify this, a pairwise comparison was performed between all 875 SQA solutions and 95 NSGA-III solutions, counting cases where one solution strictly dominates the other across all objectives. The results, shown in Table 1, show that SQA dominates NSGA-III in 93 pairs, while NSGA-III dominates SQA in 130 pairs. This indicates that both algorithms generate high-quality solutions, but they excel in different regions of the solution space.

When SQA solutions dominate, the average objective values are approximately $(F_1, F_2, F_3, F_4, F_5) \approx (-16.41, 11.30, 40.00, 105.33, 9.21)$, compared to $(-16.16, 12.50, 44.00, 111.02, 9.60)$ for the corresponding dominated NSGA-III solutions (Table 2). The largest improvements occur in sidewalk improvement cost (F_2), which decreases by roughly 9.6%, and construction cost (F_3), which decreases by approximately 9.1%. This shows that SQA is particularly effective at producing cost-focused solutions, especially in regions of the solution space with lower construction costs, by targeting solutions according to predefined weights or breakpoints.

Conversely, when NSGA-III dominates, the mean objective vector is $(-17.43, 11.84, 48.54, 134.01, 9.35)$, outperforming the dominated SQA solutions $(-17.15, 12.70, 50.00, 143.89, 10.31)$ (Table 3). The greatest improvements are seen in accident risk (F_5), reduced by $\sim 9.3\%$, and spatial dispersion (F_4), reduced by $\sim 6.9\%$. These results reflect NSGA-III's tendency to focus computational effort within high-quality Pareto neighborhoods, particularly around the $F \approx 45$ – 52 band (Fig. 6). By concentrating on the search in these promising regions, NSGA-III efficiently refines trade-offs across multiple objectives rather than evenly sampling the solution space.

Overall, these findings highlight the complementary roles of the two algorithms. SQA excels at generating operationally interpretable solutions, especially those that minimize costs, while NSGA-III is more effective at producing balanced Pareto-optimal trade-offs, improving safety, spatial coverage, and other objectives. Rather than competing for overall superiority, the two methods provide different perspectives on the solution space: SQA offers breadth and practical guidance, whereas NSGA-III provides depth and refined trade-offs. Using both together can give decision-makers a more complete understanding of the possible solutions and trade-offs in complex multi-objective optimization problems.

Table 1. NSGA-III, SQA Overview

Item	Value
SQA solutions (sqa_rows)	875
NSGA solutions (nsga_rows)	95
SQA $\rightarrow$ NSGA domination pairs (sqa_dom_nsga.count)	93
NSGA $\rightarrow$ SQA domination pairs (nsga_dom_sqa.count)	130

Table 2. SQA $\rightarrow$ NSGA (SQA dominates NSGA) — mean objective values and improvements

Metric	F_1	F_2	F_3	F_4	F_5
Dominators mean (SQA)	-16.41	11.30	40.00	105.33	9.21
Dominated mean (NSGA)	-16.16	12.50	44.00	111.02	9.60
Mean absolute difference	0.24	1.20	4.00	5.68	0.39
Mean % improvement (dominators vs dominated)	1.51%	9.57%	9.09%	5.12%	4.06%

Table 3. NSGA $\rightarrow$ SQA (NSGA dominates SQA) — mean objective values and improvements

Metric	F_1	F_2	F_3	F_4	F_5
Dominators mean (NSGA)	-17.43	11.84	48.54	134.01	9.35
Dominated mean (SQA)	-17.15	12.70	50.00	143.89	10.31
Mean absolute difference	0.28	0.87	1.46	9.88	0.96
Mean % improvement (dominators vs dominated)	1.62%	6.79%	2.92%	6.87%	9.31%

The observed two-way dominance between NSGA-III and SQA is a result of their methodological design, rather than an inconsistency in performance. SQA operates under explicit weight assignments and a fixed cardinality policy (number of sites), producing solutions that target specific operational trade-offs. For instance, SQA generates options with notably low construction costs ($F_3 \approx 40.00$) that do not appear in NSGA-III's solution set (Table 2). By contrast, NSGA-III performs an autonomous Pareto search, concentrating its search within a narrower front and refining solutions around $F_3 \approx 45 - 52$ (Fig. 6). When NSGA-III dominates, its solutions reflect this high-resolution, concentrated search; when SQA dominates, its solutions exploit cost-sensitive regions that NSGA-III underexplores.

In essence, mutual dominance arises naturally from these differing strategies: SQA deliberately samples weight-driven, operationally interpretable regions, including low- F_3 pockets, while NSGA-III intensively explores Pareto-proximal neighborhoods. Fig. 6 and Fig. 7 illustrate this effect, showing SQA's broader distribution across the objective space and NSGA-III's focused convergence. Together, they provide complementary perspectives for decision support.

To ensure a fair comparison, the runtime budgets of NSGA-III and SQA were aligned. Under this setup, NSGA-III was able to complete full convergence within the allocated time, whereas SQA distributed its runtime across multiple weight and cardinality configurations. As a result, each individual SQA execution had a reduced effective runtime. This observation suggests a practical strategy: (1) When decision-makers have clearly specified weights, dedicating the full runtime to a single SQA configuration can often yield solutions superior to NSGA-III under the same time budget. (2) Conversely, in situations without explicit preferences, NSGA-III can first be employed to autonomously converge toward balanced compromise regions, which can then be further refined using SQA under targeted weight vectors.

4.4. A Case Study of Selected Sites

Based on the results from both algorithms, we identified 40 levee-crossing site locations that meet the multi-objective optimization criteria. The weight vectors used in SQA, shown in Table 4, were generated using a layered weight scanning method. This approach systematically creates multiple sets of normalized weights (w_1, w_2, w_3, w_4, w_5) with equal step intervals of 0.025, ensuring that the total weight always satisfies $w_1 + w_2 + w_3 + w_4 + w_5 = 1$. Under the selected configuration of 40 levee-crossing sites and weight vector in Table 4, SQA produced solutions with lower objective values across all five functions compared to NSGA-III as shown in Table 5. These results indicate that, under targeted weighting, SQA can outperform NSGA-III in generating more efficient and balanced solutions within the same computational budget. This hybrid strategy, using NSGA-III for initial exploration and SQA for targeted refinement, offers a flexible and effective framework for multi-objective infrastructure planning under varying decision contexts.

Table 4. Weight of the SQA Equation

	w_1	w_2	w_3	w_4	w_5
Weight	0.375	0.250	0.125	0.125	0.125

Table 5. Comparison of SQA and NSGA-III results under the selected configuration of 40 levee-crossing sites

	F_1	F_2	F_3	F_4	F_5
SQA	-16.727	11.498	40	105.301	9.496
NSGA-III	-16.161	12.499	44	111.017	9.595

To support decision-making in levee crossing planning, spatial visualization plays a critical role in translating abstract optimization outputs into interpretable and actionable insights. By mapping selected facility locations, planners can evaluate how effectively a given configuration addresses accessibility needs, mitigates infrastructure

gaps, and promotes spatial equity. Under the selected configuration of 40 levee-crossing sites, the spatial results are derived using the SQA algorithm and weighted according to the parameters specified in Table 4. The resulting spatial outputs are visualized through three complementary maps, each corresponding to a distinct objective within the multi-objective optimization model (F_1 , F_2 , F_3 , F_4 , F_5).

4.4.1. Population Stress Relief based on the Equal Reconstruction Cost and Sidewalk Building

The spatial distribution of population stress relief reflects how effectively the selected levee-crossing sites alleviate pedestrian demand across the urban service area. Fig. 9 displays the Population Stress Relief Heatmap corresponding to the configuration of 40 selected sites. This configuration was generated through a three-objective optimization process targeting population coverage (F_1), sidewalk rebuilding needs (F_2), and site construction cost (F_3). The heatmap is derived using KDE in Equation (1), which models population demand intensity based on residential origin points. Demand relief is then calculated by tracing the shortest walking paths, represented by the sidewalk rebuilding component (F_2), from residential origins to the selected sites, under the assumption of equal reconstruction cost across all candidates (F_3). This approach allows the model to prioritize accessibility and infrastructure needs without cost bias, resulting in a spatial configuration that reflects both demand intensity and equitable service distribution. In Fig. 9, white circular markers denote the selected levee-crossing sites, while teal lines represent the shortest sidewalk rebuilding (F_2) used to evaluate both accessibility and accident risk. The color gradient along the levee encodes the degree of population stress relief, with brighter segments indicating areas where pedestrian demand is most effectively mitigated. This pattern shows that population stress relief is well distributed across the 40 selected levee-crossing sites.

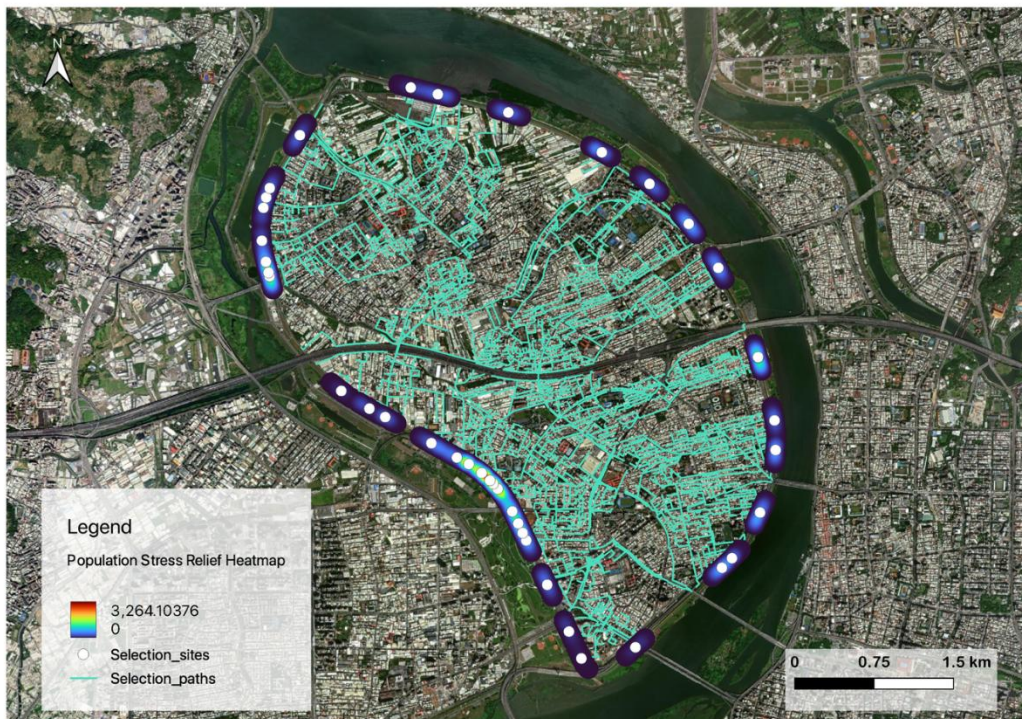


Fig. 9. Population Stress Relief Heatmap

4.4.2. Accident Risk

The optimization framework incorporates accident risk as one of its core objectives, aiming to minimize pedestrian exposure to hazardous walking conditions. This is operationalized through a NKDE of historical accident data, as defined in Equation (2). The resulting risk surface is integrated with shortest path analysis to quantify cumulative exposure along pedestrian routes connecting residential origins to selected levee-crossing sites. The underlying

heatmap in Fig. 10 visualizes traffic accident intensity across the pedestrian network, with warmer colors (red to yellow) highlighting areas of high accident concentration and cooler tones (blue to purple) indicating relatively safer zones with fewer recorded incidents. The red boundary delineates the ring levee, while white circular markers indicate the 40 selected crossing sites. This spatial representation supports risk-aware planning by identifying critical segments where infrastructure interventions may yield the greatest safety improvements.

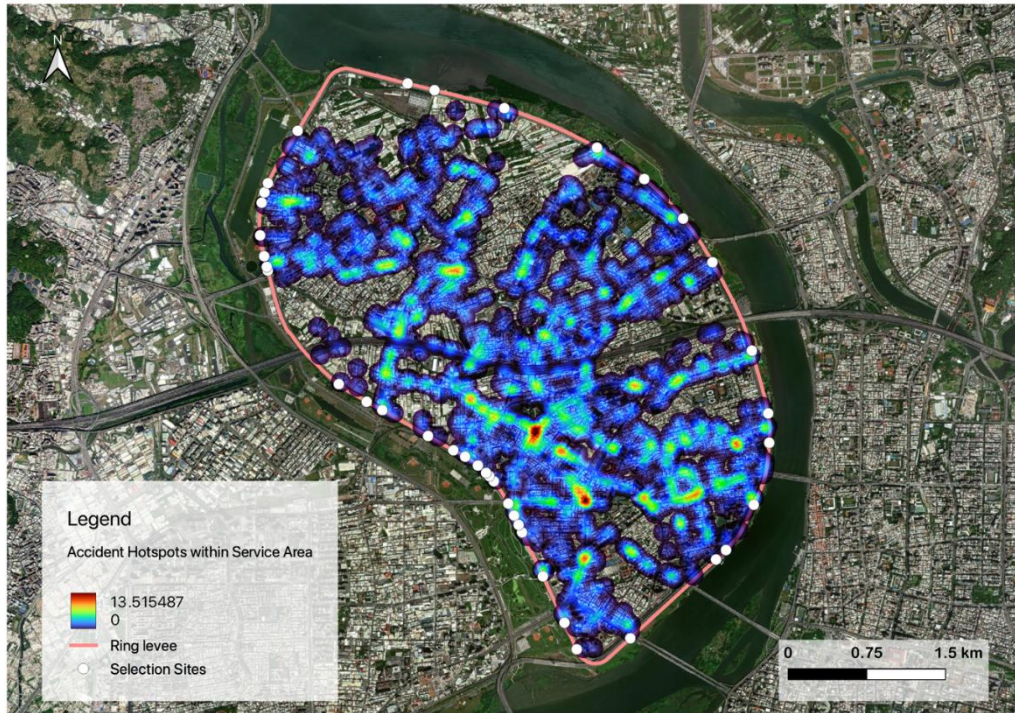


Fig. 10. Accident Hotspots within Service Area

The visualization reveals that accident hotspots are predominantly concentrated near major intersections and high-traffic corridors, particularly in zones where selected crossing sites are clustered. This spatial overlap suggests that these segments may simultaneously exhibit high accessibility demand and elevated safety risk. Such patterns are critical for interpreting the fifth objective (F_5) of the optimization model, which seeks to minimize pedestrian exposure to hazardous walking environments. By integrating accident density into the site selection process, the model promotes configurations that balance accessibility benefits with safety considerations, highlighting areas where infrastructure interventions may be most urgently needed.

4.4.3. Site Dispersion Analysis

The optimization model incorporates a dispersion penalty as Objective 4 (F_4), designed to discourage the selection of levee-crossing sites in close proximity to one another. This penalty is computed using Equation (3), which evaluates pairwise distances along the levee and embeds them into the multi-objective framework to promote a more spatially balanced configuration. Fig.11 illustrates the Cumulative Proximity Penalty along the levee, representing the spatialized outcome of the F_4 objective. The figure visualizes penalty values distributed along the levee line, where higher values (shown in red to orange) indicate segments with closely spaced selected sites, and lower values (in blue to green) reflect greater spacing between facilities. As shown, red and orange segments are concentrated in portions of the levee where selected sites are densely clustered. While these areas may correspond to zones of high pedestrian

demand, they also suggest potential redundancy in service coverage or diminished marginal accessibility gains. In contrast, blue segments highlight levee sections with wider spacing between crossings, contributing to a more equitable distribution of access across neighborhoods. This visualization provides an intuitive interpretation of the dispersion penalty, demonstrating how the model balances accessibility benefits with spatial equity considerations by discouraging excessive clustering and promoting coverage diversity.

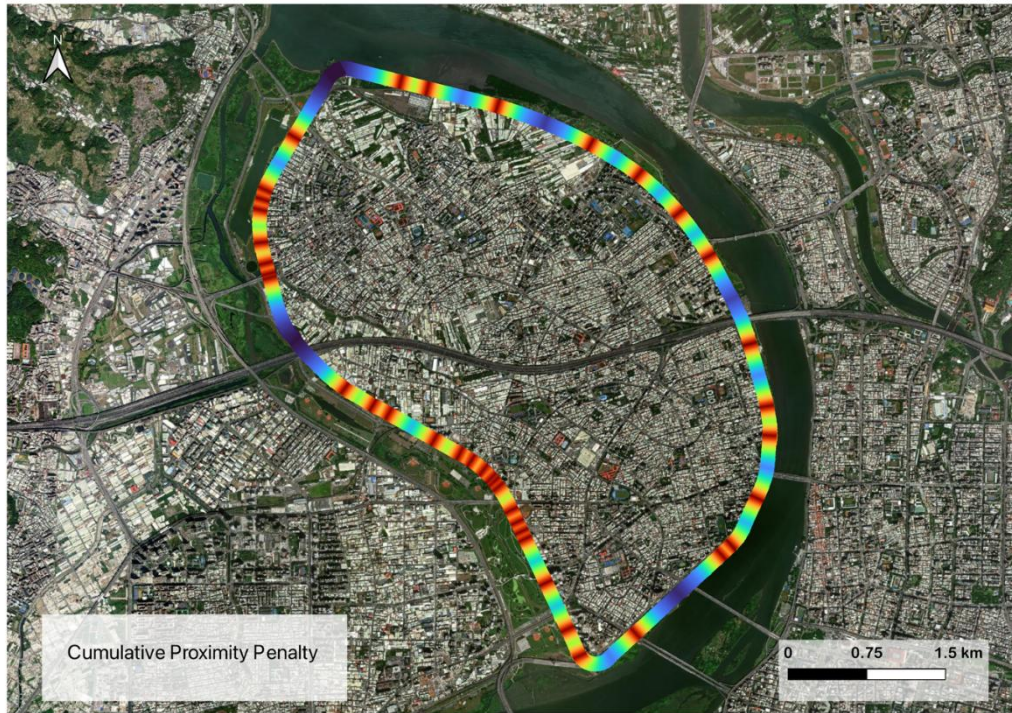


Fig. 11. Cumulative Proximity Penalty

5. Conclusion

In the context of improving connectivity across levee-adjacent urban corridors, this study's combined use of multi-objective optimization and spatial analysis provides a robust framework for levee crossing site selection and improvement. These crossings function as critical links in the pedestrian network, reconnecting neighbourhoods separated by flood-control barriers. By modelling five objectives, including population coverage, sidewalk improvement cost, construction cost, spatial equity, and accident risk, the framework captures the complex trade-offs inherent in urban infrastructure planning. NSGA-III served as a benchmark evolutionary algorithm, demonstrating reliable autonomous convergence toward balanced Pareto-proximal solutions, while Simulated Quantum Annealing (SQA) employed weight-driven scalarization and fixed cardinality constraints to produce solutions that directly reflect policy preferences and operational requirements.

Spatial analysis plays a foundational role in both preparing inputs and interpreting outputs. Before optimization begins, spatial data is used to generate all necessary indicators using methods such as KDE, network-based KDE, and exponential distance-decay functions to quantify accessibility, risk, cost, and spatial distribution. After the algorithm identifies the optimal site configuration, spatial analysis is used again to visualize selected facilities, pedestrian paths, and thematic heatmaps. This spatial interpretation ensures that the algorithm's outputs are not only technically sound but also communicable and actionable, helping planners understand where and how the selected sites deliver accessibility, safety, and equity benefits across the urban landscape.

Comparative analysis revealed that NSGA-III and SQA exhibit bidirectional dominance, reflecting their

complementary methodological designs rather than inconsistency. SQA dominates in regions emphasizing low-cost objectives, such as sidewalk improvement and construction costs, by targeting specific weight-driven solutions. Conversely, NSGA-III dominates in areas prioritizing safety and spatial equity, concentrating search effort within Pareto-proximal neighbourhoods. These patterns illustrate that SQA excels at generating operationally interpretable solutions in cost-sensitive regimes, while NSGA-III efficiently refines high-quality trade-offs across multiple objectives. Runtime considerations also inform practical deployment. Under matched budgets, NSGA-III can fully converge, whereas SQA distributes its runtime across multiple weight and cardinality combinations, resulting in reduced effective search per configuration. This suggests a sequential strategy: when decision-makers have explicit preferences, dedicating the full runtime to a single SQA configuration can yield superior solutions, whereas NSGA-III is most valuable in exploratory contexts without prescriptive weights. Applied together, NSGA-III and SQA provide a dual-method toolkit, balancing autonomous Pareto exploration with targeted, weight-driven solution generation. Both methods support informed, equitable, and implementable infrastructure decisions, highlighting the value of combining autonomous and weight-driven optimization approaches in urban planning practice.

While SQA shows promising potential, its current implementation is constrained by the lack of parallelization and tunneling effects under standalone simulations, which limits search depth and prevents it from fully surpassing NSGA-III. Additionally, SQA requires careful selection of weight vectors and cardinality settings to ensure relevant solutions, which may increase the computational burden for larger problem instances. Future research should explore GPU-based parallelization, optimized annealing schedules, and hybrid frameworks that combine NSGA-III's global convergence with SQA's weight-driven refinement. Expanding the approach to larger-scale urban networks, integrating real-time traffic or demographic data, and testing alternative quantum-inspired optimization techniques could further enhance decision-support capabilities.

Appendix A. Notation and Parameters

Table A.1. Notation and parameters for the optimization equation

Notation	Parameters
I	Set of candidate levee crossing points ($i \in I$).
J	set of population origins ($j \in J$)
W_j	population weight of origin j
d_{ij}	shortest walking distance from origin j to candidate i
h	bandwidth parameter in demand KDE
c_i	construction cost of candidate i
s_i	sidewalk improvement cost of candidate i
$H(e)$	accident intensity of network edge e (from NKDE)
L_e	length of network edge e
r_i	normalized accident risk of candidate i
λ	dispersion decay parameter
d_{ij}^{levee}	along-levee distance between candidate sites i and j
$x_i \in \{0, 1\}$	decision variable, equals 1 if a crossing is built at candidate i
w_1, w_2, w_3, w_4, w_5	weights assigned to the five objectives in the weighted-sum formulation (used in SQA).
H	Hamiltonian of the QUBO formulation, representing the energy function minimized by the annealing algorithm.

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